# v1.0Emulating Unique Electron Properties

## V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

**Architecture:** One-Electron Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
**Applies to:** `v1.0Emulating Unique Electron Properties V2.0.md`

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## Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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## V1.0 vs V2.0 Specific Changes

| Parameter | V1.0 | V2.0 | Severity | Consequence if Not Updated |
|---|---|---|---|---|
| **g-factor precision** | 9 decimal places, QED to 1-loop | 13 decimal places (CODATA 2022), sub-ppb QED | HIGH | Insufficient precision for high-fidelity electron spinor emulation |
| **Pines Demon mode** | Not present | Added — confirmed 2023 in SrVO3 (Nature 2023): massless acoustic plasmon. Requires two-band architecture | HIGH | Missing an entire observable oscillation mode of electrons in condensed phases |
| **Electron EDM** | | d_e | < 8.7×10^-29 (V1.0 era) | ACME III 2023: |
| **Compute architecture** | CPU JS particle loop, WebGL | WebGPU WGSL compute shaders (Chrome 113+ 2023); 10^7 particles vs 10^3 | HIGH | Simulation scale limited by WebGL; V2.0 WebGPU required for credible physics fidelity |
| **Berry phase** | Static scalar value | Berry tensor Omega_ij tracked dynamically per worldline fold (anomalous velocity) | MEDIUM | Static Berry phase misses topological band structure effects in multi-particle scenarios |

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## Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Compute, Pines, g-factor
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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## Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time $\tau$ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

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*End of Versioning Delta | v1.0Emulating Unique Electron Properties V1.0 archived | V2.0 is the active specification as of June 2026*